From Geographic Information System to Electrical Modeling and Analysis

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Abstract

Accurate electrical models of low-voltage (LV) distribution networks are essential for planning and operational studies under high penetration of distributed energy resources. However, manually creating such models from Geographic Information System (GIS) data is time-consuming, error-prone, and difficult to maintain. This paper proposes a systematic and automated GIS-based framework for converting multi-layer geospatial data into validated electrical network models. The methodology addresses critical challenges, including line segmentation, topology identification, feeder assignment, and transformer connectivity, while incorporating distance-based validation mechanisms to detect inconsistencies. The framework is demonstrated on a real LV network operated by the Electricity Authority of Cyprus. Results show a significant reduction in modeling time compared to manual procedures, successful feeder identification, and the ability to detect structural inconsistencies. Furthermore, the generated electrical model is used for Monte Carlo-based voltage analysis under different decentralized photovoltaic control strategies, highlighting the effectiveness of Q(V) control in mitigating overvoltage conditions with limited additional power losses compare to other control alternatives.

1 Introduction

Currently, distribution system operators (DSOs) face increasing challenges in managing and analyzing rapidly evolving electrical networks. The high integration of distributed energy resources, such as electric vehicles, heat pumps, and photovoltaics, underscores the need for large-scale analyses, including planning studies, reliability assessments, operational decision-making, and grid reinforcement [1, 2]. In addition, the control and monitoring of small-scale devices in low-voltage networks are becoming increasingly important [3]. Consequently, accurate electrical models of distribution networks are essential. At the same time, distribution networks are highly dynamic, with frequent changes in topology and connection points, making automated maintenance solutions increasingly necessary [4].

Overall, 60% of the European power system consists of low-voltage (LV) lines, 37% of medium-voltage (MV) lines, and only 3% of high-voltage (HV) lines [5]. To date, power engineers have focused on modeling this smaller, transmission-level portion of the system, typically representing the MV network in an oversimplified manner while largely neglecting the LV network, as detailed LV modeling has not been required so far. Even this limited modeling effort has taken decades and remains largely manual and time-consuming. Extending the same manual process to the distribution level would make these limitations even more pronounced, rendering such an approach impractical for large-scale, detailed studies; this likely explains why most existing works rely on a single, non-representative network.

In the literature, several approaches have been proposed to automate the generation of electrical network models. These approaches can be broadly categorized into measurement-based methods [6, 7, 8] and Geographic Information System (GIS)-based methods. In the former, the authors of [9] proposed a learning approach that exploits terminal-node voltage measurements and power flow equations to infer the network topology. Similarly, recent studies leverage smart meter measurements, combined with machine learning techniques, to capture correlations among network nodes [10]. While measurement-based approaches are particularly attractive, practical challenges, including limited measurement availability, reverse power flows that violate the monotonic power-flow assumptions, and the limited generalization capability of machine learning models under unseen operating conditions, may undermine their reliability or lead to inaccuracies during real-world operation.

On the other hand, GIS-based methodologies provide more detailed approaches by utilizing actual network drawings to derive electrical network representations. Therefore, in [4], the authors proposed a procedure to convert GIS shapefiles into Common Information Model (CIM) files that can be directly integrated into Supervisory Control and Data Acquisition (SCADA) systems. While conceptually sound, its direct implementation may lead to errors and inconsistencies, such as points located very close to each other, ignored intermediate poles, and the separate treatment of overhead lines and underground cables. Moreover, substations with multiple transformers are not explicitly addressed. In addition, a similar approach is presented in [11], where the authors introduce a plugin for electrical modeling in OpenDSS using the open-source QGIS platform. Although this work demonstrates GIS integration into an electrical simulation tool, it follows a tool-oriented approach, and the underlying methodology and handling of data

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inconsistencies are not clearly defined.

Hence, despite the advances achieved by existing modeling techniques, this work adopts a GIS-based methodology that explicitly targets these gaps, providing a general, transferable solution. In particular, the proposed methodology addresses the line segmentation problem, which is rarely formalized in the literature but is critical to avoid neglecting intermediate poles or connection points. At the same time, it ensures consistent feeder identification and unique assignments of feeders to transformers, even in substations with multiple transformers. Finally, by leveraging the data registered in the Metering Data Management System (MDMS), loads can be accurately assigned to their corresponding connection points. Therefore, the contributions of this paper are threefold:

- A systematic framework for automated extraction and processing of multi-layer GIS data, including line segmentation, node identification, and feeder assignment.
- Comprehensive distance-based validation methods for verifying connectivity between network layers and avoiding potential data inconsistencies.
- Demonstration of the methodology on real-world distribution network data, achieving reduced modeling time while improving accuracy.

The remainder of the paper is organized as follows: Section 2 presents the methodology, Section 3 describes the case study, Section 4 presents and discusses the results, and Section 5 concludes the paper.

2 Modeling Methodology

This section introduces the framework for converting GIS data into validated electrical network models, constituting the first stage of the proposed methodology. It is structured into four core substages: (i) data preprocessing and extraction, (ii) line segmentation and topology identification, (iii) feeder allocation and connectivity analysis, and (iv) model output generation. The second stage addresses electrical modeling and simulation. The overall workflow is depicted in Fig. 1.

2.1. Input Data

The proposed framework accepts three primary GIS data sources as inputs. Overhead lines are represented as *LineString* geometries describing above-ground conductors and are associated with attributes such as voltage level, conductor type, and physical length. Similarly, underground cables are modeled using *LineString* geometries to represent buried conductors with similar electrical characteristics. Transformer assets are represented as *Point* geometries indicating their geographic locations, along with attributes such as rated power and voltage-ratio information. All input data are provided in shapefile format and referenced in a common geographic coordinate system.

In addition, a mapping database is required to correctly interpret both attribute column names and their corresponding values exported from the GIS system. In practice, many GIS attributes are stored using generic or coded field names (e.g.,

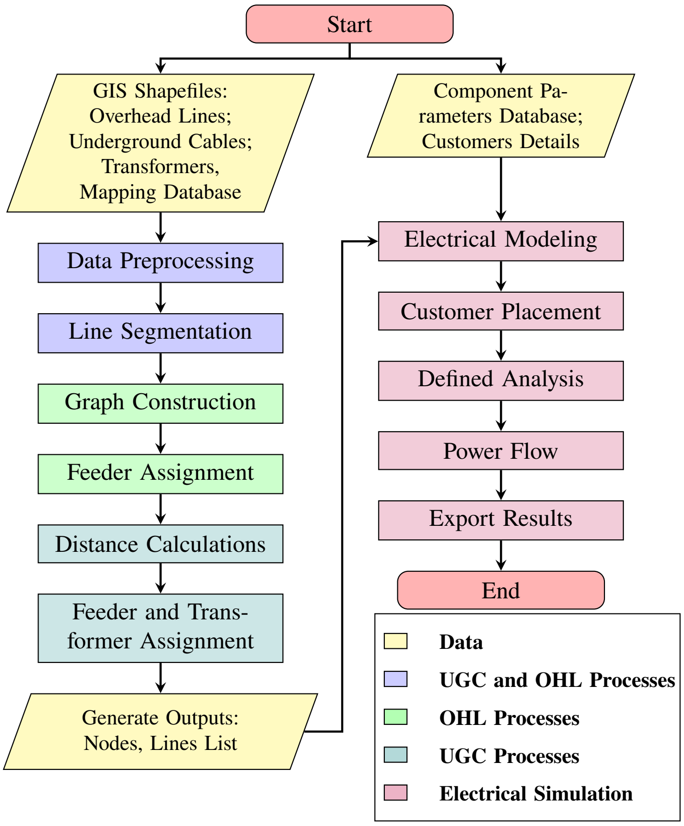

Figure 1: Framework for GIS-to-electrical model conversion.

"GIS_L_11"), which must first be mapped to their appropriate physical meanings. Furthermore, within each attribute, several electrical characteristics, such as conductor types, are encoded using internal GIS codes and must be subsequently translated into their corresponding physical descriptions. The two-stage interpretation process is depicted in Fig. 2.

A dedicated database of electrical parameters adopted by the distribution network operator is also imported, providing the necessary technical characteristics for lines and transformers. Finally, a customer database can be imported when customer information is not explicitly modeled in the GIS environment. In most practical cases, customer data is maintained in MDMS as it represents the most frequently evolving elements of the network. This database typically includes a unique identification number, installed power, the presence and rated capacity of photovoltaic

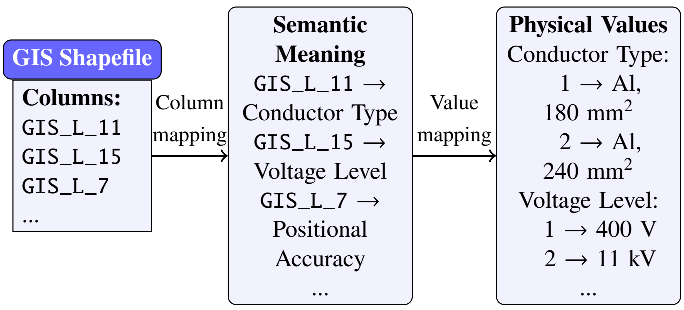

Figure 2: Two-stage interpretation of GIS attributes.

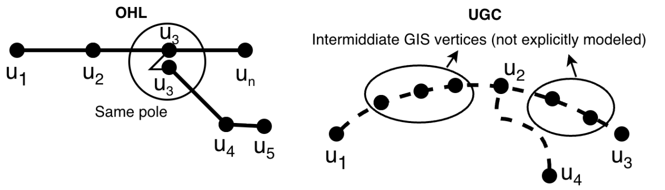

Figure 3: Illustration of the line segmentation process for overhead lines and underground cables.

(PV) systems, the applicable operating scheme, and address-based location information. Based on those inputs, customers can be assigned to the proper node, along with the corresponding details.

2.2. Data Preprocessing

The first stage involves loading and preprocessing GIS shapefiles for topological analysis. Overhead Line (OHL) LineString geometries are processed to extract their start and end coordinates, stored as tuples to facilitate spatial queries, and their geometric lengths are computed. Underground Cable (UGC) geometries follow the same procedure; however, their calculated lengths slightly underestimate the actual physical distances, as the GIS coordinate system accounts only for the horizontal (x–y) plane and ignores elevation (z). Therefore, these values are used only for validation, while the actual underground cable lengths are obtained directly from the corresponding GIS attribute field provided during the drawing stage.

2.3. Line Segmentation and Topology Identification

2.3.1. Overhead Lines Most OHL geometries in GIS databases contain multiple intermediate vertices representing the physical routing of conductors along poles. For electrical modeling purposes, these lines must be split into individual segments between consecutive vertices to accurately represent the network topology. During the segmentation process, for each LineString with n vertices $(v_1, v_2, \ldots, v_n)$, $n-1$ individual line segments are created, as defined in (1). Lines shorter than a threshold length $\ell_{\text{min}} = 2$ m are not split, as they typically represent short connection jumpers, as illustrated in Fig. 3.

$$\mathcal{L}_{seg} = \{LineString([v_i, v_{i+1}]) : i = 1, \ldots, n-1\} \quad (1)$$

where $\mathcal{L}_{seg}$ denotes the resulting set of line segments.

2.3.2. Underground Cables Underground cables share the same electrical characteristics as overhead lines; however, their intermediate vertices are typically numerous and mainly used for accurate geometric representation rather than electrical connectivity. Therefore, only junction and connection points are considered, and segmentation is applied exclusively at these locations. More specifically, for each cable endpoint, a proximity check identifies nearby intermediate vertices of other underground cables; when a match is found, the intersecting cable is split at that vertex, creating a new electrical node that may serve as a connection point for underground service lines, as illustrated in Fig. 3.

2.4. Overhead Lines Feeder Assignment

Distribution networks are typically organized into radial feeders, with sub-networks emanating from substations or distribution transformers. The overhead network is modeled as a graph $G = (V, E)$, where V denotes the set of nodes and E the set of edges. Each line segment $s \in \mathcal{L}_{seg}$ corresponds to an edge connecting the nodes $n_{\text{start}}(s)$ and $n_{\text{end}}(s)$. Based on this representation, feeders are defined as the connected components of the graph. Consequently, all line segments whose end nodes belong to the same connected component are classified as part of the same feeder, which is uniquely identified through a feeder label.

2.5. Feeder and Transformer Assignment for UGCs

A UGC may energize an overhead feeder or operate as a purely underground feeder, supplying loads through underground service connections from an energized point. To identify such connections, pairwise distances between underground cable endpoints and overhead line endpoints are computed (2). If $d_{\text{UG-OH}} < \epsilon$ (with ϵ typically set to 1 mm to account for coordinate precision), the underground cable is assigned the feeder identifier of the connected overhead line. Subsequently, a graph representation of the underground cable network is constructed in a manner analogous to that used for overhead lines. For each connected component, if at least one cable has an assigned feeder identifier, this identifier is propagated to all cables within the component. Otherwise, the remaining underground cables are classified as new underground feeders.

$$d_{UG-OH} = \min_{p_{ug} \in P_{UG}, p_{oh} \in P_{OH}} ||p_{ug} - p_{oh}||_2 \quad (2)$$

where P_{UG} and P_{OH} are the sets of endpoints of underground cables and overhead lines, respectively, and $\|\cdot\|_2$ denotes the Euclidean distance.

Finally, to assign the transformer origin to each feeder, the minimum distance is computed to all substation transformers (3). Cables with $d_{min}^{transformer} < d_{threshold}$ are assigned to those transformers. If there are several transformers, the closest one is assigned.

$$d_{min}^{transformer} = \min_{t \in \mathcal{T}} ||p_{ug} - p_t||_2 \quad (3)$$

where $\mathcal{T}$ is the set of transformers and p_t is the geographic location of transformer t.

2.6. Customer Placement

To properly define the installed power of loads, the presence of PV systems, and their rated capacity, a dedicated customer-to-network mapping procedure is applied. Specifically, a list of customers associated with the studied substation (based on postal code) is combined with address-based building location data. Using this spatial information, each customer is mapped to the nearest connection point of the distribution network, and the corresponding electrical characteristics are assigned to that node.

2.7. Output Generation

In the final stage, the processed GIS data are exported in Comma-Separated Values (CSV) format. The output includes

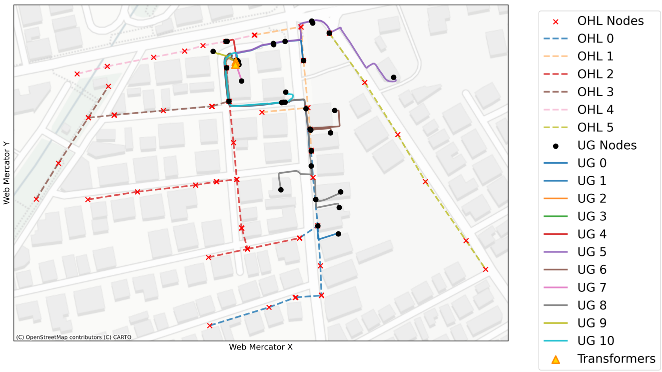

Figure 4: Geospatial representation of the studied LV network, illustrating the spatial layout of OHL, UGC, and transformers.

a node list (with node IDs, coordinates, and feeder labels) and a line list for UGCs and OHLs (with segment IDs, start and end nodes, and lengths). These files can be directly imported into any power system tool, such as Pandapower or DIgSILENT, using a custom script, together with the corresponding electrical parameters from the distribution network database, to generate the network model.

3 Case Study Description

To validate the effectiveness of the proposed methodology, it is applied to a real-world LV distribution network operated by the Electricity Authority of Cyprus in a residential area. A Monte Carlo (MC) simulation framework is employed to account for the stochastic nature of residential consumption, where load profiles are randomly assigned at each iteration. The analysis focuses on evaluating network voltage levels, with particular emphasis on overvoltage conditions observed during midday, high-PV production periods. The assessment is carried out under three different inverter control strategies, with slight parameter adjustments: fixed power factor (fixed $\cos\phi$), power factor as a function of active power ($\cos\phi(P)$), and reactive power control based on voltage ($Q(V)$).

3.1. Network Description

GIS data is extracted from the utility's GIS management system in shapefile format. The network comprises 60 overhead line segments spanning approximately 1.4 km, primarily serving aerial service drops to residential customers, and 34 underground cable segments totaling about 1.8 km (including streetlight cables) with underground service infrastructure. Within the substation, it includes two distribution transformers rated at 800 kVA and 1000 kVA, respectively, which step down the voltage from 11 kV to 400 V. The entire service area covers approximately 0.16 km^2 and supplies around 180 residential customers, with key characteristics summarized in Table 1 and the spatial layout shown in Fig. 4.

3.2. Loads and Photovoltaics

Since consumption data for the specific customers under study are unavailable, electricity consumption data from 68 residential customers in Cyprus, recorded at 30-minute intervals over 1 year, are used instead. At each simulation iteration, a random day within the study month and a random customer profile are selected and assigned to each load. In addition, due to the close geographic proximity of the households, all PV systems are assumed to follow the same generation profile, scaled by installed capacity.

4 Results and Discussion

4.1. Model Validation and Feeder Identification

The proposed framework successfully identified all network feeders. Fig. 4 illustrates the identified feeders, where UGC are represented by solid lines and OHL by dashed lines. A simplified geographic map background is overlaid for visualization purposes only, with detailed information removed to protect private data. In total, six overhead feeders were identified. Results were validated manually and through an automated comparison of assigned feeder identifiers against the *Feeder Name* attribute, confirming full consistency.

4.2. Computational Performance and Advantages

A custom ArcGIS [12] script was developed to assign substation identification numbers and export the required shapefiles. This preprocessing step requires up to 30 minutes to complete. Once the shapefiles are available, the network extraction process is computationally lightweight, taking less than 1 minute, and its cost can therefore be considered negligible. In contrast, the equivalent manual modeling process may take up to one week. Beyond the significant reduction in modeling time, the manual network modeling is inherently prone to human error and exhibits limited reproducibility, particularly because LV distribution networks evolve continuously through infrastructure upgrades, maintenance, and new customer connections.

Second, the proposed framework enables the systematic detection of network inconsistencies. This capability proved particularly valuable in the studied network, where a single overhead feeder was found to be supplied by a different transformer from that feeding the corresponding streetlight circuits installed on the same overhead line poles. Such configurations may lead to hazardous situations, for example, when maintenance personnel disconnect a feeder assuming it is de-energized, while adjacent streetlight cables remain energized from another transformer, posing a serious safety risk.

Table 1: Case Study Network Characteristics

Parameter	**OHL**	**UGC**
Number of original lines	30	30
Number of lines after segmentation	60	34
Total length (km)	1.4	1.8
Number of nodes	55	40
Number of feeders	6	11

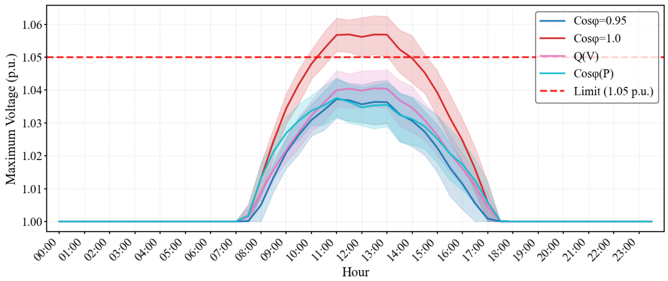

Figure 5: Maximum bus voltage for different reactive control strategies. Solid curves correspond to the mean value, and shaded regions represent the 10th–90th percentile interval.

4.3. *Voltage Analysis under a local PV Control Strategies*

Fig. 5 illustrates the resulting maximum voltage levels under the examined control strategies. The case with $cos\phi = 1$ (i.e., no reactive compensation) violates the 1.05 p.u. network limit, whereas all three local reactive control strategies maintain the voltage within acceptable bounds. As summarized in Table 2, each strategy results in different levels of reactive energy compensation and associated losses. Among them, $Q(V)$ demonstrates the most effective performance, as it adapts to real-time voltage variations, requires lower reactive power support, and limits the increase in active power losses to approximately 10%, compared to more than 20% for the other controllers; therefore, the $Q(V)$ control strategy is recommended.

Table 2: Total daily comparison of Controllers vs Baseline

Controller	Reactive Energy [MVARh]	Active Losses [MWh] (%)
$\cos\phi = 1.0$	0.0	0.51 (baseline)
$\cos\phi = 0.95$	-4.2	0.65 (+26.6%)
$Q(V)$	-1.3	0.57 (+10.0%)
$\cos\phi(P)$	-3.1	0.62 (+20.7%)

5 Conclusions

This paper presents an automated framework for converting GIS data into validated electrical models of LV distribution networks, addressing key challenges including line segmentation, feeder identification, underground–overhead connectivity, transformer assignment, and customer placement. Application to a real LV network demonstrated substantial reductions in modeling time, improved reproducibility, and effective detection of structural inconsistencies.

The resulting model enabled Monte Carlo-based power flow analysis under high PV penetration, where comparative evaluation of decentralized inverter strategies showed that Q(V) control maintains voltages within statutory limits while limiting additional active power losses. The proposed framework thus provides DSOs with a robust basis for scalable automated modeling, with future work focusing on integration with CIM and Common Grid Model Exchange Standard (CGMES) representations and real-time measurement data to enhance validation and interoperability.

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